Laboratory No.03 - Base Platform and Application Layer Protocols

Sara Viviana Arteaga Rodríguez

Julián Felipe Morales Zambrano

***Abstract*—This laboratory focused on the implementation and verification of DNS (Domain Name System) and NTP services across a heterogeneous environment. Using systems such as Slackware, NetBSD, and Windows Server, primary and secondary (slave) DNS servers were configured to ensure robust name resolution and fault tolerance. The practice also integrated Shell scripting in Linux to automate administrative tasks, including periodic task scheduling, process management through interactive menus, and file system traversal. Tools like nslookup and dig were utilized to validate the accuracy and reliability of the configured services.**

I. Objetivo

Continuar aprendiendo la instalación del software base, en particular los servicios DNS y NTP, complementado con conocimientos de programación en Shell.

II. Introducción

En las redes empresariales modernas, la correcta configuración de los servicios de infraestructura es esencial para garantizar la estabilidad, la conectividad y la disponibilidad de las aplicaciones. Entre estos servicios, el Sistema de Nombres de Dominio (DNS) es uno de los componentes más críticos.

El DNS permite a los usuarios y sistemas traducir nombres de host descriptivos en direcciones IP legibles por máquinas, lo que facilita una comunicación fluida entre recursos distribuidos.

Esta práctica de laboratorio se centra en la implementación de servidores DNS en un entorno heterogéneo, utilizando diferentes sistemas operativos como Solaris, Slackware, CentOS y Windows Server. Trabajar con múltiples plataformas proporciona a los estudiantes una perspectiva más amplia sobre cómo implementar, configurar y probar un mismo servicio en diversas condiciones.

También se enfatiza la verificación y la resolución de problemas utilizando herramientas como nslookup y dig.

Finalmente, el ejercicio integra scripts de shell y gestión de procesos para fortalecer las habilidades de automatización.

III. 1. Linux DNS Server - BIND

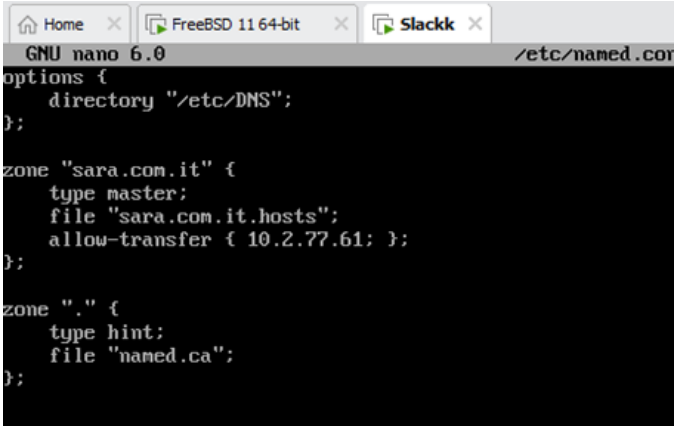
En la máquina virtual Slackware descargamos BIND.

```
root@Sara:~# mount /dev/cdrom /mnt/cdrom
mount: /mnt/cdrom: WARNING: source write-protected, mounted read-only.
root@Sara:~# cd /mnt/cdrom/slackware/n
root@Sara:/mnt/cdrom/slackware/n# installpkg bind*.txz
Verifying package bind-9.16.25-i586-1.txz.
Installing package bind-9.16.25-i586-1.txz [REC]:
PACKAGE DESCRIPTION:
# bind (DNS server and utilities)
#
# The named daemon and support utilities such as dig, host, and
# nslookup. Sample configuration files for running a simple caching
# nameserver are included. Documentation for advanced name server
# setup can be found in /usr/doc/bind-9.x.x/.
#
Executing install script for bind-9.16.25-i586-1.txz.
Package bind-9.16.25-i586-1.txz installed.
root@Sara:/mnt/cdrom/slackware/n# umount /mnt/cdrom
umount: /mnt/cdrom: target is busy.
root@Sara:/mnt/cdrom/slackware/n# cd /
root@Sara:/# umount /mnt/cdrom
root@Sara:/# ls /var/log/packages | grep bind
root@Sara:/# which named
/usr/sbin/named
root@Sara:/# named -v
BIND 9.16.25 (Extended Support Version) <id:3e14423>
root@Sara:/# which nslookup
/usr/bin/nslookup
root@Sara:/# mkdir /etc/DNS
root@Sara:/# ls /etc
DIR_COLORS              exports             minirc.dfl          rc_maps.cfg
DNS/                    exports.d/          misc@               request-key.conf
HOSTNAME                fb.modes            mke2fs.conf         request-key.d/
ImageMagick-7/          file/               mkinitrd.conf.sample resolv.conf
ModemManager/           fonts/              modprobe.d/         resolvconf.conf
NetworkManager/         fstab               motd                rmt@
OpenCL/                 ftpusers            mplayer/            rndc.key
UPower/                 fuse.conf           mtab@               rpc
X11/                    gamin/              mtools.conf         samba/
a2ps-site.cfg           gconf/              mutt/               sane.d/
a2ps.cfg                genpowerd.conf      my.cnf              sasl2/
acpi/                   glmp/               my.cnf.d/           screenrc
aliases                 gnupg/              mysqlaccess.conf    sdda.conf
aliases.db              gpm-root.conf       nail.rc             securetty
alsa/                   gpm-syn.conf        named.conf          security/
anthy-conf              gpm-twiddler.conf   nanorc              sensors.d/
asciidoc/               group               ndctl/              sensors3.conf
```

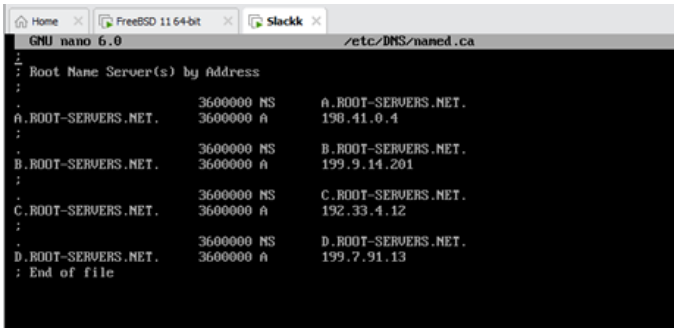
Creamos los archivos DNS named.conf, named.ca y sara.com.it.

```
root@Sara:/# mkdir /etc/DNS
mkdir: cannot create directory '/etc/DNS': File exists
root@Sara:/# cd /etc/DNS
root@Sara:/etc/DNS# pwd
/etc/DNS
root@Sara:/etc/DNS# nano /etc/DNS/named.ca
root@Sara:/etc/DNS# nano /etc/named.conf
root@Sara:/etc/DNS# vi /etc/named.conf
root@Sara:/etc/DNS# nano /etc/named.conf
root@Sara:/etc/DNS# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:d8:1b:cc brd ff:ff:ff:ff:ff:ff
    inet 10.2.77.56/24 brd 10.2.77.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fed8:1bcc/64 scope link
       valid_lft forever preferred_lft forever
root@Sara:/etc/DNS# mkdir -p /etc/DNS
root@Sara:/etc/DNS# cd /etc/DNS
root@Sara:/etc/DNS# nano /etc/DNS/santiago.com.it.hosts
root@Sara:/etc/DNS# nano /etc/DNS/sara.com.it.hosts
root@Sara:/etc/DNS# cat /etc/named.conf
options {
    directory "/etc/DNS";
};

zone "sara.com.it" {
    type master;
    file "sara.com.it.hosts";
    allow-transfer { 10.2.77.59; 10.2.77.61; };
};

zone "." {
    type hint;
    file "named.ca";
};
```

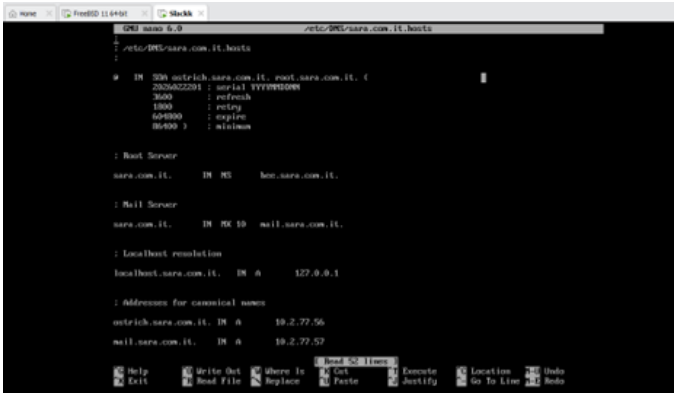
Contenido del archivo /etc/named.conf.



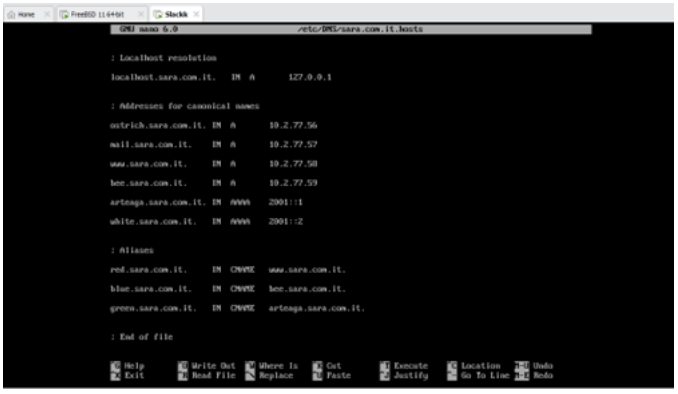
Contenido del archivo /etc/DNS/named.ca.



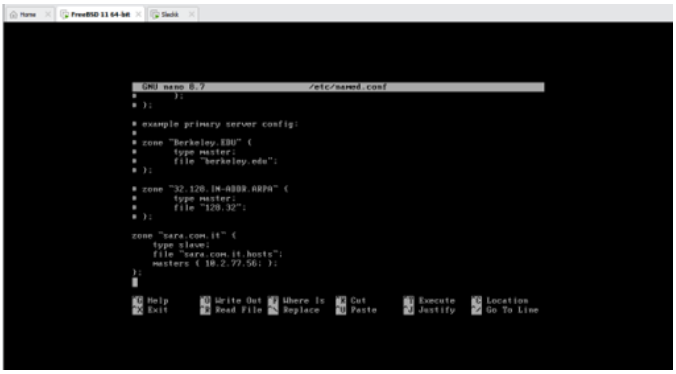
Contenido del archivo /etc/DNS/sara.com.it.hosts.



Contenido del archivo /etc/named.conf en NetBSD.



Observamos el proceso de configuración y puesta en marcha del servidor DNS BIND en el servidor Slackware. Primero, accedimos al directorio /etc/DNS, donde se encuentran los archivos de configuración como named.ca y el archivo de zona sara.com.it.hosts, correspondiente al dominio configurado como primario. Luego se inicia el servicio ejecutando el proceso named (el servidor DNS), y finalmente se verifica su funcionamiento utilizando el comando ps aux | grep named, confirmando que el proceso está activo y que el servidor DNS se encuentra operando correctamente.

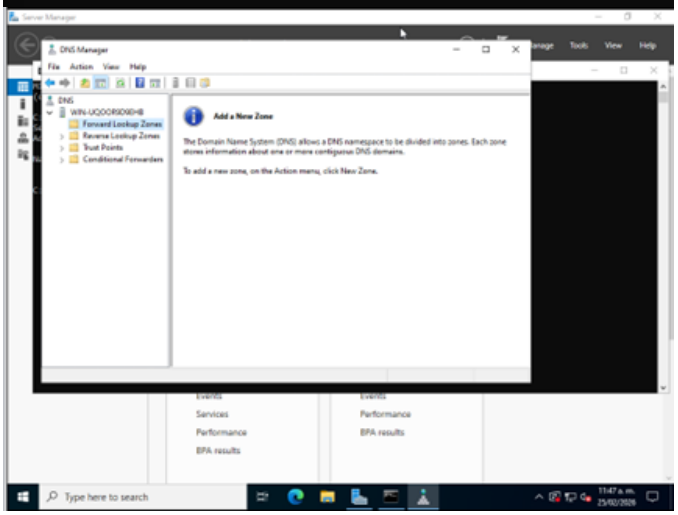


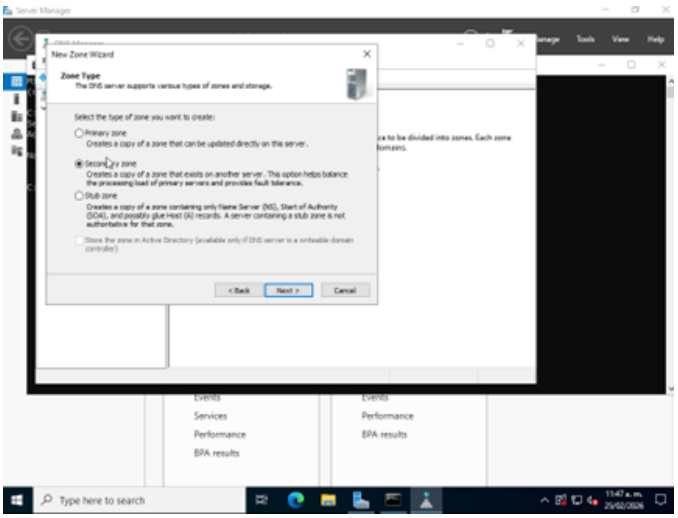
Al ejecutar el comando nslookup, verificamos el correcto funcionamiento del servidor DNS configurado. Con esta prueba comprobamos que el servidor es capaz de resolver un nombre de dominio y traducirlo a su dirección IP correspondiente.

```
root@Sara:/etc/DNS# ls /etc/DNS
named.ca  sara.com.it.hosts
root@Sara:/etc/DNS# nano /etc/DNS/named.ca
root@Sara:/etc/DNS# named
root@Sara:/etc/DNS# /etc/rc.d/rc.bind start
-su: /etc/rc.d/rc.bind: Permission denied
root@Sara:/etc/DNS# chmod +x /etc/rc.d/rc.bind
root@Sara:/etc/DNS# /etc/rc.d/rc.bind start
Starting BIND:  /usr/sbin/named -u named
root@Sara:/etc/DNS# ps aux | grep named
root      1275  0.1  0.4  77140 18784 ?        Ssl  09:32   0:00 named
root      1301  0.0  0.0   3864  2004 pts/0    S+   09:32   0:00 grep named
```

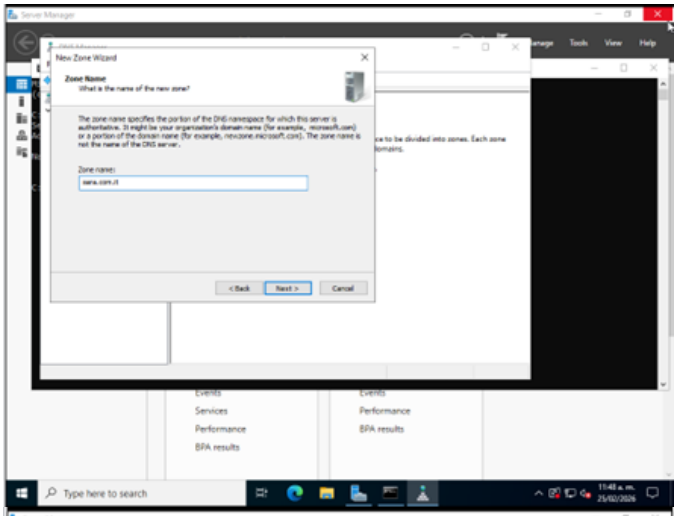
En Windows Server, para el tipo de instalación, elija la opción "Instalación basada en funciones o roles".

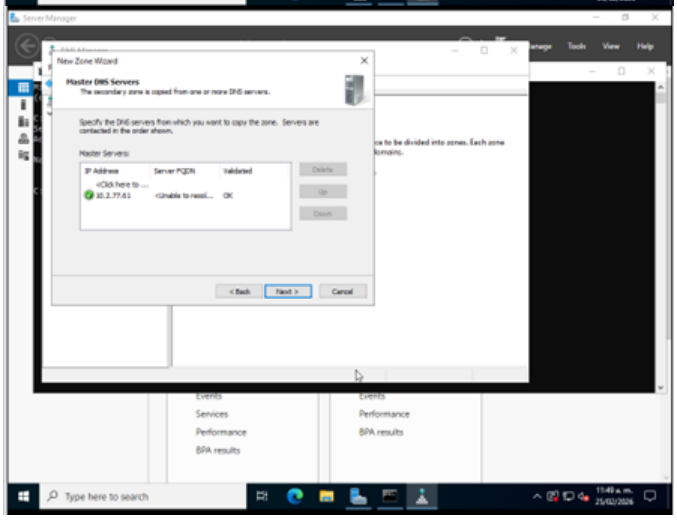
```
root@Sara:/etc/DNS# nslookup www.sara.com.it 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   www.sara.com.it
Address: 10.2.77.58

root@Sara:/etc/DNS# nslookup mail.sara.com.it 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   mail.sara.com.it
Address: 10.2.77.57

root@Sara:/etc/DNS# nslookup bee.sara.com.it 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   bee.sara.com.it
Address: 10.2.77.59

root@Sara:/etc/DNS# nslookup ostrich.sara.com.it 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   ostrich.sara.com.it
Address: 10.2.77.56

root@Sara:/etc/DNS# nslookup red.sara.com.it 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

red.sara.com.it canonical name = www.sara.com.it.
Name:   www.sara.com.it
Address: 10.2.77.58

root@Sara:/etc/DNS#
```



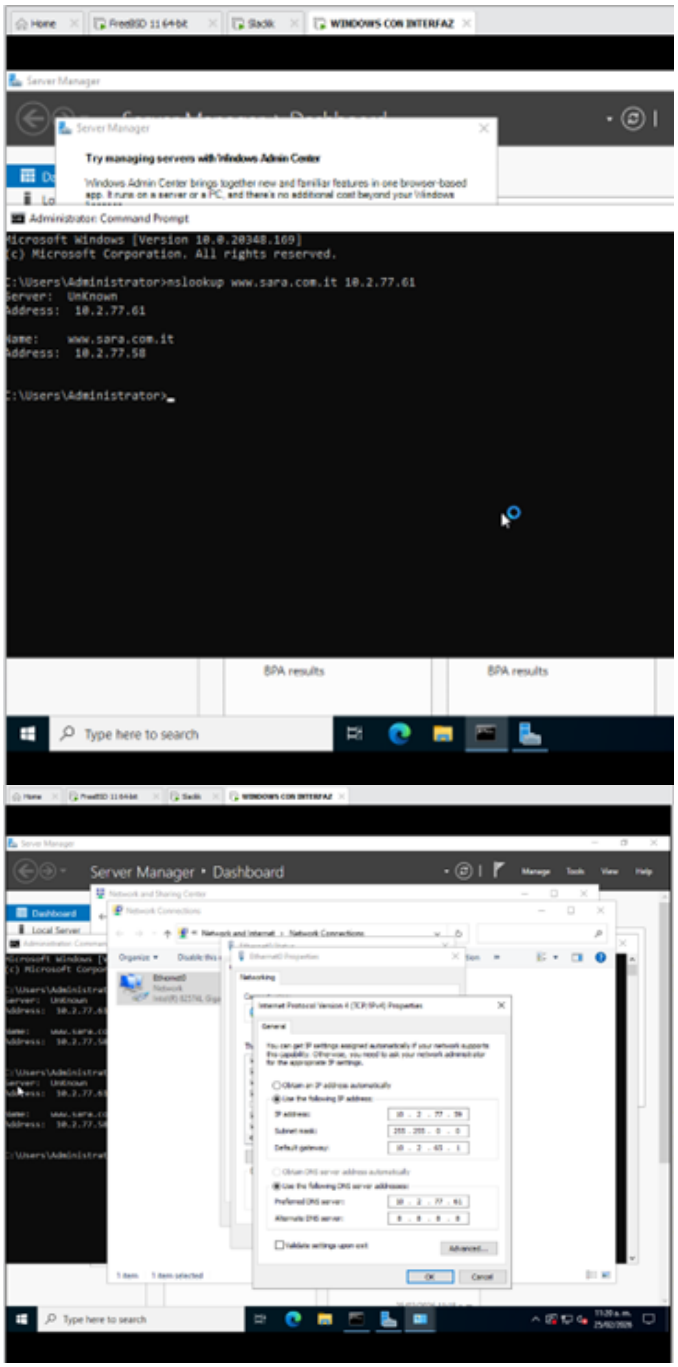


Después de terminar con la instalación y la configuración, usamos nslookup para probar que el servidor esclavo esté ejecutándose.

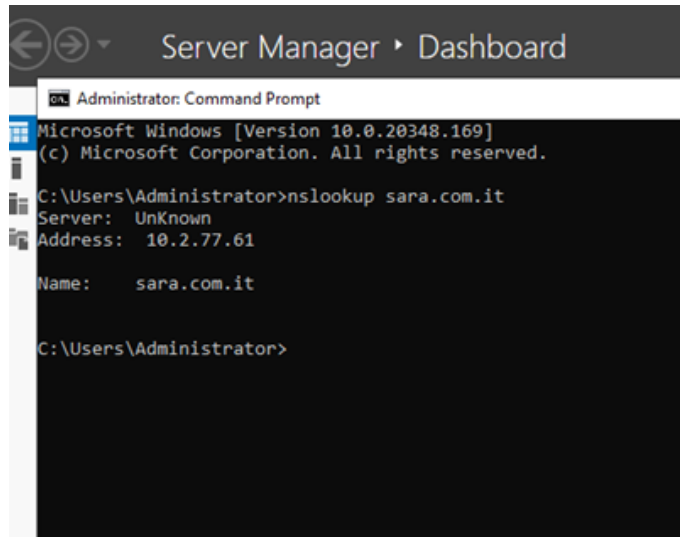




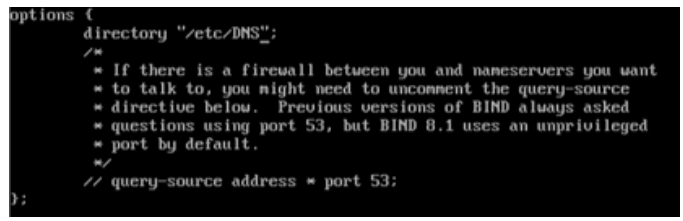
En NetBSD configuramos el servicio para julian.org.uk.



En Slackware configuramos el DNS esclavo para el segundo dominio y añadimos la zona de julian.org.uk.



Revisamos la conexión del esclavo usando nslookup.

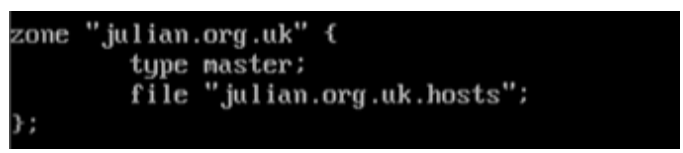


IV. 2. Other Useful Commands

En Slackware:

a. Allow configuring a task to run periodically on the system. The user will specify the task to be executed and its frequency via the command line. The parameters should NOT be prompted interactively.

Nombre del script: scheduleTaskScript.sh.



b. Build a Shell with a menu of options, where one option is to exit, and the others execute the desired command and return to the options menu. The menu should allow displaying the processes currently running, searching for a process, killing a process, and restarting a process.

Nombre del script: processMenu.sh.

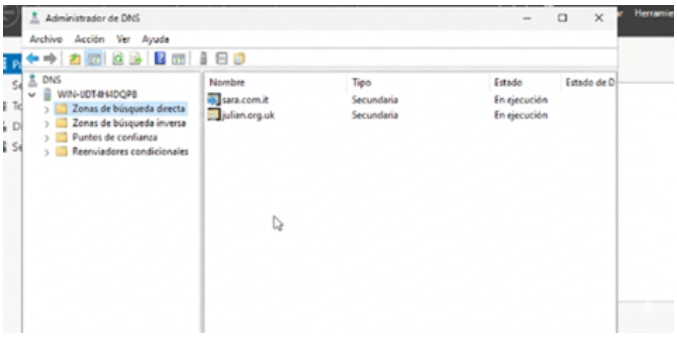
```
; Root Name Server(s) by Address
;
                        3600000         NS      A.ROOT-SERVERS.NET.
A.ROOT-SERVERS.NET.     3600000         A       198.41.0.4
;
                        3600000         NS      B.ROOT-SERVERS.NET.
B.ROOT-SERVERS.NET.     3600000         A       199.9.14.201
;
                        3600000         NS      C.ROOT-SERVERS.NET.
C.ROOT-SERVERS.NET.     3600000         A       192.33.4.12
;
                        3600000         NS      D.ROOT-SERVERS.NET.
D.ROOT-SERVERS.NET.     3600000         A       199.7.91.13
; End of file_

;       /etc/DNS/julian.org.uk.hosts file
;
;
;       INCLUDE UPDATE SOA HEADER
@ IN SOA snake.julian.org.uk.   root.julian.org.uk. (
        2025090901 ; Serial
        3600 ; Refresh
        1800 ; Retry
        604800 ; Expire
        86400 ; Minimum TTL
)
;
; Root server
;
julian.org.uk.          IN      NS      pocillos.julian.org.uk.
;
; Mail server
;
julian.org.uk.          IN      MX      10      mail.julian.org.uk.
;
; Localhost resolution
;
localhost.julian.org.uk.        IN      A       127.0.0.1
;
; Addresses for canonical names
;
snake.julian.org.uk.    IN      A       10.2.77.32
;
mail.julian.org.uk.     IN      A       10.2.77.34
www.julian.org.uk.      IN      A       10.2.77.35
pocillos.julian.org.uk. IN      A       10.2.77.31
blackmouth.julian.org.uk.       IN      AAAA    2001::1
caracoles.julian.org.uk.        IN      AAAA    2001::2
;
; Aliases
```

c. Create a Shell that allows traversing the file system from a given directory, including subdirectories, and shows the n smallest files within a size specified by the user. The output should indicate: file name, path, and size.

Nombre del script: filesScript.sh.

```
orange.julian.org.uk.   IN      CNAME   www.julian.org.uk.
white.julian.org.uk.    IN      CNAME   pocillos.julian.org.uk.
gray.julian.org.uk.     IN      CNAME   blackmouth.julian.org.uk.
;
; End of file
zone "julian.org.uk" {
        type slave;
        file "slaves/db.julian.org.uk";
        masters { 10.2.77.56; };
};
```

```
root@Sara:~# nslookup pocillos.julian.org.uk
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   pocillos.julian.org.uk
Address: 10.2.77.31
```




```
root@Sara:~# nano scheduleTaskScript.sh
root@Sara:~# chmod +x scheduleTaskScript.sh
root@Sara:~# cat /tmp/cron-test.txt
cat: /tmp/cron-test.txt: No such file or directory
root@Sara:~# ./scheduleTaskScript.sh * * * * *
Task scheduled successfully:
filesScript.sh processMenu.sh scheduleTaskScript.sh filesScript.sh processMenu.sh date >> /tmp/cron-te
st.txt
root@Sara:~#
```

```
root@Sara:~# nano scheduleTaskScript.sh
root@Sara:~# chmod +x scheduleTaskScript.sh
root@Sara:~# ./scheduleTaskScript.sh "*/5 * * * *" "date >> /tmp/test.txt"
Task scheduled successfully.
root@Sara:~# nano processMenu.sh
root@Sara:~# chmod +x processMenu.sh
root@Sara:~# ./processMenu.sh
------ MENU ------
1. Show running processes
2. Search process
3. Kill process
4. Restart process
5. Exit
------------------
Choose option: 1
  PID COMMAND         %MEM %CPU
  117 kworker/1:1-eve  0.0  0.1
  208 jbd2/sda2-8      0.0  0.1
  601 udevd            0.1  0.1
    1 init             0.0  0.0
    2 kthreadd         0.0  0.0
    3 rcu_gp           0.0  0.0
    4 rcu_par_gp       0.0  0.0
    6 kworker/0:0H-ev  0.0  0.0
    7 kworker/0:1-eve  0.0  0.0
------ MENU ------
1. Show running processes
2. Search process
3. Kill process
4. Restart process
5. Exit
------------------
Choose option:
```

```
root@Sara:~# nano filesScript.sh
root@Sara:~# chmod +x filesScript.sh
root@Sara:~# ./filesScript.sh 10 1G
./.gnupg/pubring.gpg~ 0
./.gnupg/secring.gpg 0
root@Sara:~#
```

```
# nano scheduleTaskScript.sh
# chmod +x scheduleTaskScript.sh
# ./scheduleTaskScript.sh * * * * *
"*":18: bad minute
crontab: errors in crontab file, can't install.
Task scheduled successfully:
scheduleTaskScript.sh scheduleTaskScript.sh scheduleTaskScript.sh scheduleTaskScript.sh scheduleTaskSc
ript.sh date >> /tmp/cron-test.txt
# ./scheduleTaskScript.sh "*" "*" "*" "*" "*"
Task scheduled successfully:
* * * * * date >> /tmp/cron-test.txt
#
```

```
# nano processMenu.sh
#  chmod +x processMenu.sh
# ./processMenu.sh
------ MENU ------
1. Show running processes
2. Search process
3. Kill process
4. Restart process
5. Exit
------------------
Choose option: 1
USER  PID %CPU %MEM    VSZ   RSS TTY   STAT STARTED    TIME COMMAND
root    0  0.0  1.4      0 57340 ?     DWl   2:26PM 0:01.74 [system]
root    1  0.0  0.0  19956  1576 ?     Is    2:26PM 0:00.02 init
root  416  0.0  0.1  37724  2512 ?     Ss    2:26PM 0:00.13 /usr/sbin/syslogd -s
root  700  0.0  0.1  70552  3276 ?     Is    2:26PM 0:00.00 sshd: /usr/sbin/sshd [listener] 0 of 10-1
root  714  0.0  0.2 341040 10332 ?     Is    2:26PM 0:00.02 /usr/pkg/sbin/smbd -D
root  726  0.0  0.2 339576  7292 ?     I     2:26PM 0:00.01 smbd: notifyd
root  728  0.0  0.2 339568  6936 ?     S     2:26PM 0:00.01 smbd: cleanupd
root  730  0.0  0.1 271924  4308 ?     Ss    2:26PM 0:00.46 /usr/pkg/sbin/nmbd -D
root  819  0.0  0.0  24584  1476 ?     Is    2:26PM 0:00.00 /usr/sbin/inetd -l
root  881  0.0  0.0  19788  1388 ?     Is    2:26PM 0:00.00 /usr/sbin/powerd
sara  895  0.0  0.1  85328  5608 ?     S     7:32PM 0:02.49 sshd: sara@pts/0 (sshd)
root  937  0.0  0.0  20320  1704 ?     Is    2:26PM 0:00.01 /usr/sbin/cron
root  944  0.0  0.2  85188  7652 ?     Is    7:32PM 0:00.01 sshd: sara [priv]
root 1118  0.0  0.4 138100 14944 ?     Isl   7:46PM 0:00.54 /usr/sbin/named
------ MENU ------
1. Show running processes
2. Search process
3. Kill process
4. Restart process
5. Exit
------------------
Choose option: |
```

```
# nano filesScript.sh
# chmod +x filesScript.sh
```

```
# ./filesScript.sh 5 10000
# find / -type f -size -10000k 2>/dev/null | head
# find / -type f 2>/dev/null | head
/etc/fstab
/etc/mtree/set.modules
/etc/mtree/set.base
/etc/mtree/NetBSD.dist
/etc/mtree/set.etc
/etc/mtree/special
/etc/mtree/set.comp
/etc/mtree/set.games
/etc/mtree/set.gpufw
/etc/mtree/set.man
# nano filesScript.sh
# nano filesScript.sh
# ./filesScript.sh 5 1
File: /etc/dumpdates
Size: 0 bytes
----------------------
File: /etc/skeykeys
Size: 0 bytes
----------------------
File: /proc/1578/cmdline
Size: 0 bytes
----------------------
File: /proc/1578/environ
Size: 0 bytes
----------------------
File: /proc/1578/limit
Size: 0 bytes
----------------------
# |
```

V. Conclusiones

- La configuración de servidores maestros y esclavos es una estrategia crítica para proporcionar redundancia, balanceo de carga y tolerancia a fallos en la resolución de nombres.
- El uso de herramientas de diagnóstico como nslookup permitió confirmar que la traducción de nombres de dominio a direcciones IP funciona correctamente tras la configuración de las zonas.
- La integración de Shell scripting facilita la administración de sistemas al automatizar procesos repetitivos, como la gestión de tareas programadas y la supervisión de procesos activos.